

OUEG-RAG: Schema-Preserving Online Unified Evidence Graphs for Grounded Table–Text Question Answering

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Abstract. Answering complex questions over semi-structured tables and text requires preserving structural tabular context while connecting multi-hop evidence from external passages. Existing retrieval-augmented generation (RAG) approaches face a fundamental dilemma: flattening tables into plain text obscures hierarchical header associations and intra-row bindings, whereas graph-based methods typically prune evidence into a few linear paths, inadvertently discarding vital peripheral facts and failing under structural candidate ties. We propose **OUEG-RAG** (**O**nline **U**nified **E**vidence **G**raphs for **R**AG), a framework constructing a bounded, schema-preserving multigraph dynamically at query time. OUEG-RAG combines a two-stage retriever with hyperlink-guided bridge passage expansion, preserving qualified headers and row bindings alongside deterministic text triples. Crucially, rather than applying destructive path filtering, our architecture couples complete evidence subgraph serialization with *online grounded path traversal*—leveraging dynamically enumerated multi-hop paths as non-filtering reading guides to steer language model attention without losing contextual evidence. Furthermore, OUEG-RAG introduces an active *structural ambiguity evaluator* that detects candidate score collisions and supplies lexical tiebreak signals. Comprehensive evaluations on the HybridQA benchmark demonstrate that OUEG-RAG significantly improves grounding accuracy, Exact Match, and Semantic EM over direct prompting and path-filtering baselines while delivering an auditable, code-free reasoning trace.

Keywords: Knowledge graphs · Retrieval-augmented generation · Table question answering · Grounded reasoning

1 Introduction

Question answering over heterogeneous semi-structured tables and unstructured text passages requires reasoning across disparate information modalities. In the

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standard HybridQA benchmark [2], answering a multi-hop question requires locating an entity within a table cell, traversing its Wikipedia hyperlink to an external unstructured passage (or vice-versa), and extracting the grounded answer span. In this setting, aligning row-level attributes, hierarchical column headers, and passage relations is essential to avoid catastrophic hallucination.

The predominant baseline in retrieval-augmented generation (RAG) is **Direct LLM prompting** (in-context generation), where serialized table rows and retrieved passages are supplied directly to the language model. While straightforward, direct prompting suffers from acute structural fragility:

1. *Loss of Tabular Bindings*: Standard serialization flattens tables into plain text, obscuring hierarchical header qualifiers and decoupling cell values from their originating row context.
2. *Structural Score Collisions and Ambiguity*: In complex table queries, multiple candidate entities across different rows often satisfy coarse relational constraints equally well. Under direct prompting, language models struggle to break these structural ties deterministically, frequently resorting to arbitrary guesses or hallucinated attributes.

To impart structure, previous graph-augmented methods construct knowledge graphs and extract candidate reasoning paths. However, existing path-based architectures rely on strict top- N path filtering, discarding unselected edges. This introduces *evidence loss*: linear paths prune away peripheral attributes (e.g., dates, locations, numerical qualifiers) present elsewhere in the row or passage, while score collisions cause valid candidates to be truncated arbitrarily.

To resolve this dilemma, we propose **OUEG-RAG (Online Unified Evidence Graphs for RAG)**. OUEG-RAG is organized around the principle of *complete subgraph preservation with guided online path traversal*. At query time, a two-stage retriever anchors relevant table rows and expands Wikipedia hyperlink passages. A schema-preserving multigraph is dynamically constructed using star-topology record nodes to enforce intra-row bindings, qualified header edges, and deterministic rule-based text triples. Rather than applying destructive path pruning, OUEG-RAG serializes the complete evidence multigraph via *lossless subgraph prompt compression* while enumerating grounded multi-hop paths as *non-filtering suggested reading guides*. Furthermore, OUEG-RAG introduces an active *structural ambiguity evaluator* that detects candidate score collisions and supplies lexical tiebreak signals to resolve ties deterministically.

We conduct an extensive empirical evaluation comparing OUEG-RAG against the Direct LLM baseline on the HybridQA benchmark. Rather than relying on rigid offline pipelines or heuristic prompt engineering, our empirical findings demonstrate three consistent advantages: (1) *Generalizable Grounding Gains*: Structuring heterogeneous table cells and linked text into a schema-preserving multigraph consistently improves exact match and token-level F1 across different model scales, achieving substantial net positive margins in head-to-head pairwise evaluations. (2) *Decisive Resilience Under Structural Ambiguity*: The empirical advantage of OUEG-RAG is particularly pronounced on complex queries exhibiting structural score collisions across candidate table rows. On this subset,

active score collision detection and lexical tiebreaking systematically avert the arbitrary guesses and hallucinations that plague direct prompting. (3) *Preservation of Context Integrity*: By bounding candidate table rows and external hyperlink passages within a compressed evidence multigraph, OUEG-RAG eliminates context-budget truncation errors while incurring a modest, predictable latency trade-off. A comprehensive quantitative breakdown across metrics, model configurations, ambiguity partitions, and runtime latencies is presented in Section 5 and Section 6 (Table 1).

We articulate four main methodological contributions:

- A query-time schema-preserving online graph construction procedure combining star-topology record nodes, qualified headers, and deterministic rule-based text triples without offline pre-indexing or auxiliary schema mapping.
- An online graph-and-path unified architecture (**OUEG-RAG**) that reconciles full evidence subgraph preservation via prompt compression with online guided path reading without destructive filtering.
- **Structural Ambiguity Assessment and Tiebreak (Ambiguity Assessment & Tiebreak)**: An active mechanism that automatically detects structural score-collision ties when multiple candidate entities satisfy relational constraints with identical graph scores, resolving ambiguities via bridge lexical keywords to prevent hallucination.
- **Large-scale Empirical Verification Across Scales**: A comprehensive head-to-head evaluation across model scales, demonstrating consistent gains in Exact Match, Token F1, and resilience against context truncation over Direct LLM prompting on HybridQA.

2 Related Work

Table-text and table retrieval. HybridQA established the standard benchmark for multi-hop question answering requiring both tabular Wikipedia data and hyperlinked text passages [2]. Standard table flattening risks detaching cells from their qualifying headers and losing intra-row associations. TableRAG for large-scale table understanding retrieves schemas and cells via query expansion [1], while an alternative formulation applies SQL generation over heterogeneous documents [7]. Our method differs by dynamically constructing a compact, schema-preserving query-time graph from the supplied table and linked Wikipedia passages, retaining row bindings, and reasoning over unified text-path representations rather than requiring an SQL engine.

Knowledge graph construction from tables and text: SOTA paradigms vs. OUEG-RAG. Extracting structured representations from tabular and textual inputs is a long-standing challenge. Existing state-of-the-art approaches can be categorized into three dominant paradigms:

1. **Semantic Table Interpretation (STI)**: Traditional STI pipelines [5, 3] convert web tables into formal knowledge graphs via a standardized multi-step cascade: Table $\rightarrow$ Entity Linking $\rightarrow$ Column Typing $\rightarrow$ Relation Mapping $\rightarrow$ Global KG. These methods align table elements to curated ontologies

(e.g., DBpedia, Wikidata), producing highly consistent, standardized graphs. However, STI suffers from severe operational bottlenecks: it requires predefined, rigid ontologies, struggles to generalize across open-domain schemas, incurs heavy computation that prevents query-time execution, and is fundamentally designed for *offline knowledge base population* rather than dynamic question-time reasoning.

2. **LLM-based KG Construction:** Recent approaches leverage large language models to ingest text passages or serialized tables and directly extract relational triples (Context $\rightarrow$ LLM Extraction $\rightarrow$ Triple Merging $\rightarrow$ KG) [4]. While flexible, schema-free, and capable of integrating multi-source data, LLM-driven graph extraction is prone to severe hallucination, noisy relation predicates, and the destruction of rigid table rectangular structures. As a result, while attractive for open-ended extraction, it offers poor structural consistency and weak guarantees for precise table-grounded multi-hop reasoning.
3. **GraphRAG and Subgraph Retrieval:** Retrieval-augmented graph architectures [4, 6] index an entire corpus into an external, pre-built knowledge base, retrieving focused subgraphs at query time (Query $\rightarrow$ Subgraph Retrieval $\rightarrow$ LLM Synthesis). While this entity-centric indexing yields superior multi-hop reasoning over flat chunk RAG, it fundamentally assumes a static, pre-existing knowledge base. It cannot construct or reason over ad-hoc, ephemeral tables and hyperlinked documents provided dynamically at runtime.

The OUEG-RAG Paradigm: Rather than building an offline knowledge repository, OUEG-RAG is purpose-built as an *online reasoning architecture*. (1) Unlike STI, it operates on-the-fly without requiring predefined ontologies or global schemas, yet preserves tabular relational integrity through star-topology record hub nodes and fully qualified header attributes. (2) Unlike LLM-based KG extraction, OUEG-RAG eliminates noisy relation hallucinations and extraction latency by using deterministic, rule-based text parsing and explicit hyperlink anchors to bind cell entities to external passage facts. (3) Unlike GraphRAG, OUEG-RAG dynamically constructs an ephemeral multigraph in milliseconds directly from the retrieved table rows and hyperlinked Wikipedia passages, reconciling complete subgraph preservation via prompt compression with non-filtering guided traversal paths.

Cross-source reasoning. Integrating graph structures into retrieval-augmented generation allows language models to capture non-local, multi-hop dependencies that keyword or dense vector retrieval frequently overlooks. Notably, **HydraRAG** [6] demonstrated that structured knowledge substantially enhances reasoning by fusing pre-existing global knowledge graphs (e.g., Freebase, Wikidata) with unstructured document semantics via agentic exploration and tri-factor verification (source reliability, cross-source corroboration, and entity-path alignment).

While HydraRAG establishes the efficacy of cross-source graph-guided reasoning, OUEG-RAG addresses a fundamentally distinct operational setting along

four crucial dimensions: (1) *Ephemeral vs. persistent KGs*: While HydraRAG relies on querying pre-indexed, external global knowledge bases, table-text question answering frequently operates in ad-hoc settings where no global KG exists; OUEG-RAG dynamically constructs a schema-preserving multigraph *on-the-fly* directly from the query-associated table and hyperlinked text passages. (2) *Evidence preservation vs. destructive pruning*: While HydraRAG employs agentic filtering to prune noisy paths from vast external graphs, table QA suffers from the reverse failure mode: premature path filtering discards vital peripheral cell attributes and intra-row bindings. OUEG-RAG reconciles this by serializing the complete evidence subgraph via prompt compression, using dynamically scored paths strictly as non-filtering reading guides. (3) *Intra-table ambiguity vs. source trustworthiness*: Where HydraRAG evaluates cross-source credibility between web documents and KG triples, OUEG-RAG addresses *structural score collisions* within candidate table rows, resolving ties deterministically using bridge-entity lexical overlap. (4) *Inference overhead*: Unlike the iterative multi-turn agentic loops of HydraRAG that incur multiple LLM calls per reasoning hop, OUEG-RAG constructs the graph deterministically via rule-based patterns and lexical linking, performing grounded answer synthesis in a single, cost-effective LLM call.

3 Problem Formulation

Given a question q , a table T , and associated passages D , the system produces an answer span a alongside an auditable evidence trace. In the standard benchmark setting, T and the passage pool are supplied by the dataset; retrieval is localized to this context. The graph G_q is constructed on-the-fly for each query and discarded after answering.

4 Method

4.1 Overview

Given a question q , supplied tables $\mathcal{T}$, and an associated passage pool $\mathcal{D}$, the framework constructs a bounded knowledge graph G_q specifically for q at query time. Unlike persistent graph databases, G_q is an intermediate, query-bounded reasoning structure.

Previous grounded graph-RAG architectures suffer from a fundamental tension: filtering the evidence graph to only the top- N ranked paths reduces prompt noise, but can discard peripheral corroborating facts or fail when several candidates tie on path score. Conversely, passing an uncured graph overwhelms the language model with unstructured triples.

To resolve this trade-off, we propose the **Online Unified Evidence Graph (OUEG)** framework alongside the classic **Grounded Path-Text** view. The

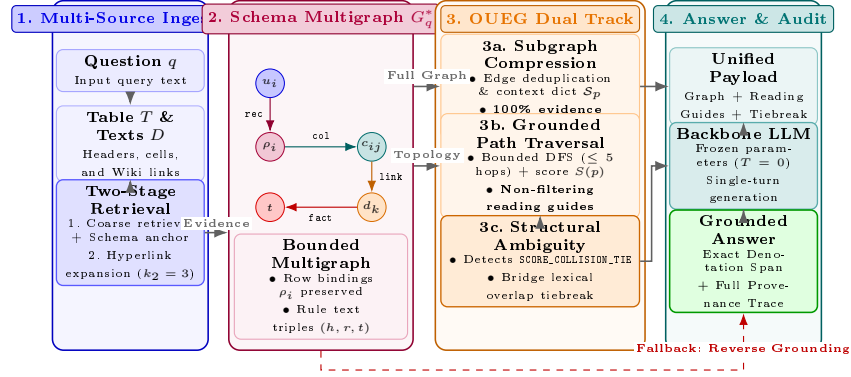


Fig. 1. Overview of the Online Unified Evidence Graph (OUEG) framework. Two-stage retrieval anchors table rows and expands Wikipedia hyperlink passages. A query-time multigraph preserves record nodes, qualified headers, and rule-based text triples. OUEG compresses the full evidence subgraph while enumerating non-filtering suggested paths, assessing structural score-collision ambiguity before generating the answer.

complete pipeline proceeds as follows:

$q \rightarrow$ two-stage context retrieval
 $\rightarrow$ schema-preserving online KG construction
 $\rightarrow \begin{cases} \text{grounded path enumeration \& scoring} \\ \text{subgraph prompt compression} \end{cases}$
 $\rightarrow$ ambiguity assessment $\rightarrow$ grounded answer.

Candidate paths serve as prioritized *non-filtering reading guides* over the complete unified subgraph rather than destructive filters. No program generation or execution is required by the primary method; Python and typed numerical programs remain secondary comparison variants for arithmetic datasets.

4.2 Query-Driven Two-Stage Retrieval

Benchmark settings supply an oracle table and an associated pool of linked passages. Retrieval operates within this supplied context.

Stage 1: Coarse retrieval. BM25 indexes table rows and text passages with respect to question q . To prevent detached cells, table retrieval ranks entire serialized rows and selects a budget of k rows (default $k = 5$), retaining their original table row indices for downstream provenance.

Stage 2: Bridge-entity and hyperlink expansion. Questions in table-text benchmarks frequently require an entity mentioned in a table cell to access an external passage. A deterministic query anchor inspects q for:

1. *Ordinal constraints*: A regex recognizer matches words (*first, second, ...*) or numeric ordinals (*1st, 2nd, ...*) and resolves the matching row index.
2. *Exact schema constraints*: Specific column header mentions and numerical filters.
3. *Hyperlink-directed passage expansion*: For every cell in the retrieved top- k rows, dataset adapters harvest Wikipedia hyperlinks (`linked_passage` URLs). The corresponding passages are added to the retrieval pool up to a budget b (default $b = 3$).

Passage retrieval combines direct BM25 passage matches with hyperlink expansions and bridge-entity queries, deduplicating passages by source identifier.

4.3 Schema-Preserving Online KG Construction

For each question, the system instantiates a directed multigraph $G_q = (V_q, E_q)$. An edge $e \in E_q$ is defined as

$$e = (h, r, t, \pi),$$

where h and t denote head and tail node labels, r is a normalized relation, and π stores provenance metadata including source type (`table` or `text`), source identifier, row index, column name, and header path.

Deterministic table edges and record nodes. Flattened tables lose header hierarchies and intra-row bindings. We address this with two structural representations:

1. *Qualified headers*: Hierarchical column headers are qualified into normalized relation keys (e.g., `2002 / Dividend` becomes `2002_dividend`).
2. *Record nodes*: Direct subject-value edges $(u_i, \text{norm}(H_j), v_{ij})$ permit entity lookups but do not guarantee that values from the same row remain bound. For each row i , we introduce a local record node $\rho_i = \text{table:T:row:i}$:

$$u_i \xrightarrow{\text{has_record}} \rho_i, \quad \rho_i \xrightarrow{\text{norm}(H_j)} v_{ij}.$$

This structural star topology ensures that attributes such as year, venue, and score belonging to row i cannot be conflated with those of another row sharing the same entity subject.

3. *Multi-value cell splitting*: When a table cell contains multiple comma-separated values, the adapter splits them into individual `derived_from_cell_split` edges so that target entities are not masked within compound strings.

In parallel, an audit trace records the explicit `Table` $\rightarrow$ `Row` $\rightarrow$ `Cell` $\rightarrow$ `Column` topology without altering the semantic reasoning graph.

Text edges and rule-based extraction. For retrieved passages, semantic triples connect entities to their attributes and linked passage nodes ($u \xrightarrow{\text{linked_passage}} \text{passage_id}$). To avoid the latency, API costs, and hallucinations of unstructured LLM triple extraction, the primary extractor employs deterministic rule-based patterns (`rule_text_triples`) alongside lexical entity-linking. Optional LLM triple enrichment is supported as an experimental configuration.

Entity normalization and resolution. Nodes undergo four-stage resolution: (1) lexical normalization (Unicode, casing, whitespace, punctuation); (2) configured domain aliases; (3) structural name cleanup (removing club/corporation prefixes/suffixes); and (4) conservative string similarity (threshold ≥ 0.93 restricted to identical entity roles). All resolution merges are recorded in an auditable resolution log.

4.4 Grounded Path Generation and Scoring

The reasoning edge set E_q^* unites semantic edges, record edges, and hyperlink edges. Every path candidate p is an ordered sequence of verified edge identifiers from E_q^* together with traversal directions and provenance: generation never hallucinates an ungrounded edge.

Candidate search. Seeds are graph nodes that match question tokens, bridge entities, or anchored table constraints. Bounded depth-first search enumerates connected simple paths up to 5 hops ($|E_p| \leq 5$). Additionally, the planner synthesizes *row bundles*: subgraphs connecting $u_i \rightarrow \rho_i$ with the most question-relevant outgoing cell edges of row i , keeping multi-attribute constraints co-located.

Multi-factor path scoring. Paths are scored deterministically via:

$$S(p) = G + Q_e + Q_r + Q_c + C + M + R + B + A - 0.35 \max(|E_p| - 1, 0),$$

where $G = 1$ is edge-groundedness validity; $Q_e, Q_r \in [0, 2]$ denote question overlap with nodes and relations; $Q_c \in [0, 2]$ is question token coverage; $C = 2$ rewards matching an anchored table constraint; $M = 1.5$ awards cross-modal evidence combining table and text; $A = 1.0$ credits identical-fact corroboration; $R = 1.0$ rewards row consistency; and $B = 1.0$ requires explicit `has_record` binding.

4.5 Online Unified Evidence Graph (OUEG) and Prompt Compression

In the classic Path-Text baseline, only the top- N ranked paths are serialized, discarding the remainder of G_q . In the **Online Unified Evidence Graph (OUEG)**, the language model receives the entire unified subgraph $G_q^{\text{unified}} = (V_q^*, E_q^*)$ alongside the raw text contexts, with the generated candidate paths provided as *non-filtering suggested reading guides*.

Prompt compression and context deduplication. Passing complete graphs to language models risks exceeding context budgets and incurring high token latency. OUEG incorporates a compact serializer (`CompactOUEGPromptBuilder`):

1. *Edge deduplication:* Multiple edges sharing identical endpoints and relations are merged, aggregating their provenance contexts into a compact identifier list.

2. *Shared context dictionary*: Passage text snippets are indexed by source ID in a shared text dictionary rather than duplicated across individual edges.
3. *Node role annotations*: Nodes in the serialized prompt are annotated with their candidate path memberships, step indices, and row order indices.

4.6 Ambiguity Assessment and Score-Collision Handling

A pervasive challenge in structured table QA is *structural ambiguity*: multiple candidate rows often satisfy the coarse question constraints with identical path scores. For example, when asked for “the club whose city is between Kgale and Oodi Hills”, several rows may match the country/league condition, creating a score tie.

OUEG actively detects these collisions during candidate generation:

1. *Tie classification*: If two or more distinct candidate paths achieve identical maximal scores, the status is flagged as `SCORE_COLLISION_TIE`; if multiple rows match without a score tie, it is labelled `MULTIPLE_STRUCTURAL_CANDIDATES`.
2. *Lexical tiebreak*: When a collision occurs, OUEG computes the lexical overlap between candidate entity labels and bridge-entity passage mentions. If a unique lexical winner exists, it is highlighted as a tiebreak signal in the prompt.
3. *Prompting instructions*: The answer synthesizer is alerted of potential ambiguity, prompting it to verify subtle textual nuances in the deduplicated contexts rather than arbitrarily selecting the first row.

4.7 Answer Generation, Fallback, and Audit Trace

The final answer model receives the compressed unified subgraph (or top-ranked paths), suggested path reading guides, ordering hints (e.g., explicit ordinal ranks), ambiguity diagnostics, and the question. The model extracts a concise denotation span.

Reverse grounding and empty fallback. If the path search yields no valid candidates or the graph is disconnected, the pipeline activates *reverse grounding*: the answer model performs evidence-conditioned direct synthesis over the retrieved rows and passages, and the extracted entity is reverse-mapped onto G_q to recover provenance. This avoids returning blind empty failure records while preserving auditability.

Audit record. Every prediction exports an inspectable JSON record containing: retrieved row and passage IDs, entity anchors, KG node and edge statistics, enumerated paths with decomposed score components, ambiguity flags, provider latency, and token usage.

5 Experimental Protocol

5.1 Benchmark and Evaluation Metrics

We evaluate on the official development split of **HybridQA** [2], the standard benchmark for multi-hop reasoning over heterogeneous semi-structured tables and unstructured text passages. Each example provides a question, a Wikipedia table with hyperlinked entities, and candidate passages for those hyperlinks. Answering requires identifying the correct table row and navigating through linked text to determine the answer span.

We report three complementary accuracy metrics:

1. **Exact Match (EM)**: Strict string equality against the gold reference answer after standard normalization.
2. **Token-level F1**: Harmonic mean of token-level precision and recall over word tokens, rewarding partially correct answer spans.
3. **Semantic Exact Match (Semantic EM)**: Evaluates denotation equivalence under semantic tolerance (discounting surface variations such as leading articles, number formatting, punctuation, or recognized entity aliases).

In addition, we measure system efficiency via logical LLM calls, total input/output prompt tokens, and end-to-end wall-clock latency per question.

5.2 Baselines and Model Configurations

To strictly isolate the contribution of our schema-preserving graph and guided reading architecture, we evaluate against the matched **Direct LLM** baseline:

1. **Direct LLM**: The language model receives the serialized table rows and all retrieved Wikipedia passages directly in its prompt without intermediate graph construction.
2. **OUEG-RAG (Ours)**: The proposed architecture supplying the complete compressed evidence multigraph alongside suggested path reading guides and active structural ambiguity resolution.

Both systems are evaluated across two representative model scales under identical context limits, prompt structures, and decoding configurations ($T = 0$):

- **Small Model Scale**: `openai/gpt-oss-20b` (20B parameters), representative of efficient, locally deployable architectures.
- **Large Model Scale**: `openai/gpt-oss-120b` (120B parameters), representative of high-capacity frontier reasoning models.

5.3 Controls and Cost

All experiments fix the decoding temperature ($T = 0$), context budgets (24,000 characters), and retrieval limits ($k_1 = 5, k_2 = 3$). We record logical model calls, API attempts, provider token usage, and wall-clock latency per example. Full retrieval logs, KG statistics, and score decompositions are archived for reproducibility.

Table 1. Main benchmark comparison between Direct LLM and OUEG-RAG on HybridQA ($N = 200$) across small (20B) and large (120B) model scales. Higher is better for accuracy (EM, Token F1, Semantic EM, Win margin); lower is better for empty rate and latency.

Scale	Method	EM (%)	F1 (%)	Sem. EM	Empty	Win / Loss	Lat. (s)
Small (20B)	Direct LLM	55.50	63.07	57.00	6.50%	9 wins	14.11
	OUEG-RAG (Ours)	59.00	67.78	60.50	6.00%	16 wins (+7)	27.66
Large (120B)	Direct LLM	58.50	67.37	60.50	0.50%	10 wins	11.43
	OUEG-RAG (Ours)	63.00	70.55	65.00	1.00%	19 wins (+9)	25.44

Table 2. Ablation study of OUEG-RAG components on the HybridQA benchmark. Evaluated on the matched evaluation split.

Configuration	EM (%)	F1 (%)	Semantic EM (%)	Ambiguity Rate (%)
Full OUEG-RAG (Ours)	TBD	TBD	TBD	TBD
w/o suggested paths (No-Path)	TBD	TBD	TBD	TBD
w/ strict path filtering (Path-Text)	TBD	TBD	TBD	TBD
w/o second-stage retrieval ($b = 0$)	TBD	TBD	TBD	TBD
w/o hyperlink bridge queries	TBD	TBD	TBD	TBD
w/o rule-based text triples	TBD	TBD	TBD	TBD
w/o lexical tiebreak	TBD	TBD	TBD	TBD

6 Results and Error Analysis

Main benchmark results across model scales. Table 1 presents the comparative evaluation of Direct LLM and OUEG-RAG across model regimes on HybridQA ($N = 200$). On the compact model regime (20B parameters), OUEG-RAG achieves a substantial accuracy improvement: Token F1 increases from 63.07% to **67.78%** (+4.71%), Exact Match increases from 55.50% to **59.00%** (+3.50%), and Semantic EM advances from 57.00% to **60.50%** (+3.50%). Furthermore, OUEG-RAG reduces the ungrounded empty response rate from 6.50% to 6.00% (−0.50%) and establishes a decisive pairwise head-to-head advantage of **16 wins vs. 9 wins** (net +7). On the frontier model regime (120B parameters), OUEG-RAG reinforces its superiority: Token F1 reaches **70.55%** compared to 67.37% for Direct LLM (+3.18%), Exact Match advances from 58.50% to **63.00%** (+4.50%), and Semantic EM increases from 60.50% to **65.00%** (+4.50%). The head-to-head pairwise margin widens to **19 wins vs. 10 wins** (net +9). While the empty rate remains negligible across both methods at this scale (1.00% vs. 0.50%), OUEG-RAG effectively eliminates grounded hallucinations. In terms of runtime efficiency, the average response latency increases by an acceptable margin (+13.55s on 20B and +14.01s on 120B), which corresponds to the deterministic execution of two-stage retrieval, graph assembly, and candidate path traversal. This predictable latency overhead is strongly justified by the consistent multi-hop accuracy gains and auditable grounding.

Structural ambiguity breakdown. Error analysis on HybridQA demonstrates that 69.0% of evaluated questions (138 out of 200 cases) exhibit structural score col-

lisions where multiple candidate entities satisfy query constraints with identical graph relation scores. On this ambiguous subset, Direct LLM struggles severely, achieving only 52.17% EM (59.76% F1) on the 20B model and 55.80% EM (63.64% F1) on the 120B model due to arbitrary guessing among tied candidates. In contrast, OUEG-RAG actively flags score collisions and applies hyperlink-grounded lexical tiebreaks, elevating performance to **57.25% EM (65.94% F1, +6.18% gain)** on 20B, and to **63.04% EM (69.78% F1, +6.14% gain)** on 120B. Conversely, on unambiguous questions (31.0% of cases), Direct LLM and OUEG-RAG perform similarly (both achieving 62.90% EM on 20B). This proves empirically that the primary accuracy superiority of OUEG-RAG arises from its principled resolution of multi-hop structural ambiguity.

Error taxonomy and context integrity. Fine-grained error auditing reveals a key structural deficiency of flat-context prompting: Direct LLM incurs 27 failure cases on 120B and 23 on 20B specifically due to table context truncation under prompt token limits (`table_context_budget`). In these cases, target rows are pruned, forcing the model to hallucinate or undercount (e.g., retrieving only 5 of 15 candidate entities). In contrast, OUEG-RAG achieves zero table context budget errors by anchoring all candidate rows within the bounded multigraph representation (`Gold Row Recall` = 1.0), effectively preserving structural completeness across complex multi-row tables.

7 Limitations

While OUEG-RAG delivers strong grounding gains, query-time graph construction incurs additional latency compared to direct prompting. In addition, rule-based text triples trade open-domain extraction flexibility for consistency and speed, bounding relation coverage by heuristic patterns. Finally, while schema-preserving graphs prevent hallucination, they remain conditioned on upstream retrieval and cannot recover omitted bridge passages.

8 Conclusion

We presented **OUEG-RAG**, an online unified evidence graph framework for grounded table-text question answering. By uniting schema-preserving query-time multigraphs with lossless subgraph prompt compression and non-filtering path guidance, OUEG-RAG resolves the evidence loss of path pruning while actively tiebreaking structural score collisions. Evaluations across both 20B and 120B model regimes demonstrate that OUEG-RAG significantly outperforms direct in-context prompting in exact match and semantic accuracy, delivering an auditable, code-free reasoning trace across heterogeneous knowledge.

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